

1-8B Llama models || best among 2 LRs || Tasks: Im_eval (left), JRT (middle), multi-hop (right)

8B 1.0T(4096) Llama - original - lr(0.002,0.003)	66.9%	66.4%	81.2%	78.6%	71.1%	83.3%	56.0%	48.3%	80.9%	92.0%	46.2%	39.0%	65.5%	60.6%	9.60	4.60	64.7%	81.4%	89.0%	73.4%	81.1%	42.8%	45.7%	54.7%	69.2%	74.4%	73.6%	34.8%	56.4%	100.0%	99.9%	99.4%	97.4%	95.0%	88.8%	2.6%	0.6%	54.1%	37.2%	33.1%	30.9%	32.3%	32.4%	1.7%	0.7%
8B 1.0T(4096) Llama - Canon -ABCD - lr(0.002,0.003)	67.7%	69.4%	80.6%	79.5%	72.9%	84.1%	57.8%	46.7%	83.5%	89.0%	47.4%	39.5%	67.4%	62.4%	9.19	4.20	64.8%	82.2%	85.4%	75.9%	74.1%	43.0%	47.2%	54.6%	71.0%	74.9%	73.6%	36.0%	59.5%	100.0%	99.9%	99.7%	98.7%	98.2%	94.6%	26.7%	9.7%	50.8%	42.5%	37.0%	34.5%	35.1%	33.9%	10.0%	4.5%
3B 1.0T(4096) Llama - original - lr(0.002,0.003)	62.3%	63.8%	79.6%	74.3%	65.9%	79.4%	50.3%	44.9%	73.5%	88.0%	44.8%	38.9%	52.7%	53.2%	11.45	5.47	58.1%	77.6%	88.1%	59.3%	63.8%	38.8%	41.7%	52.0%	62.3%	72.2%	68.6%	30.3%	42.6%	100.0%	99.6%	98.2%	94.7%	89.3%	54.6%	0.0%	0.2%	35.2%	36.4%	34.5%	33.9%	30.3%	30.5%	0.1%	0.1%
3B 1.0T(4096) Llama - Canon -ABCD - lr(0.002,0.003)	63.8%	64.9%	79.1%	75.0%	67.8%	79.8%	51.6%	45.5%	76.7%	87.0%	45.0%	38.4%	62.7%	56.1%	11.09	5.09	59.8%	81.8%	87.9%	63.5%	65.7%	41.0%	44.4%	53.3%	65.1%	71.9%	67.3%	30.9%	45.1%	100.0%	100.0%	99.6%	97.8%	96.8%	93.6%	70.2%	10.0%	37.2%	36.8%	32.8%	34.6%	31.3%	32.8%	24.6%	6.6%
1.3B 2.0T(4096) Llama - original - lr(0.003,0.005)	57.7%	59.2%	77.2%	69.3%	63.6%	76.6%	44.8%	44.7%	69.1%	82.0%	40.4%	39.0%	38.6%	45.5%	13.30	7.03	56.4%	81.5%	86.4%	59.3%	62.6%	36.7%	37.2%	49.0%	61.5%	69.1%	66.8%	26.4%	40.5%	100.0%	98.7%	96.1%	93.7%	90.9%	57.1%	0.1%	0.2%	34.2%	30.2%	27.4%	29.7%	27.3%	13.2%	0.0%	0.0%
1.3B 2.0T(4096) Llama - Canon -ABCD - lr(0.003,0.005)	58.7%	59.1%	78.1%	70.2%	65.0%	76.9%	47.0%	43.9%	69.9%	82.0%	41.4%	37.9%	44.8%	46.8%	12.86	6.86	57.7%	80.9%	86.0%	60.9%	61.8%	39.0%	41.6%	51.4%	62.6%	69.0%	66.1%	29.2%	43.7%	100.0%	99.5%	98.8%	98.7%	96.6%	90.8%	23.3%	15.1%	35.1%	33.6%	33.2%	32.0%	28.0%	31.1%	27.4%	7.4%
1.3B 1.0T(4096) Llama - original - lr(0.002,0.003)	57.7%	57.5%	76.9%	68.7%	64.0%	77.0%	46.0%	44.4%	69.4%	81.0%	41.8%	38.1%	41.4%	44.5%	13.67	7.54	56.6%	78.8%	86.6%	60.7%	60.6%	36.4%	38.9%	48.5%	63.1%	68.4%	66.5%	28.2%	42.3%	100.0%	99.5%	96.8%	92.1%	72.7%	39.8%	0.1%	0.1%	33.2%	30.2%	30.6%	30.6%	29.8%	24.9%	0.0%	0.0%
1.3B 1.0T(4096) Llama - Canon -ABCD - lr(0.002,0.003)	58.9%	59.4%	77.0%	69.5%	63.6%	77.4%	46.7%	44.2%	71.2%	86.0%	40.8%	37.8%	45.5%	47.0%	13.17	7.07	57.8%	80.9%	85.6%	61.7%	63.7%	38.4%	42.0%	51.2%	64.7%	69.1%	65.2%	29.0%	42.2%	100.0%	99.8%	99.6%	99.4%	95.7%	93.1%	39.4%	7.6%	34.4%	34.6%	33.8%	32.8%	30.8%	30.8%	27.8%	6.1%
Im_eval(avg13)		LMB	PIQA	Hella	Wino	ARC-e	ARC-c	SIQA	BoolQ	COPA	OBQA	RACE	CSQA	MMLU	Wiki ppl	LMB ppl	JRT(avg12)	FDA	FDA2	SWDE	SWDE2	NQ	NQ2	Squad	Squad2	TriviaQA	TriviaQA2	Drop	Drop2	1-hop-0k	1-hop-0.5k	1-hop-1k	1-hop-2k	1-hop-3k	1-hop-4k	1-hop-5k	1-hop-6k	2-hop-0k	2-hop-0.5k	2-hop-1k	2-hop-2k	2-hop-3k	2-hop-4k	2-hop-5k	2-hop-6k